TOF Project Assistant

Challenge B

Accountable AI Project Management Assistant | AI CAN DO IT Hackathon Indonesia 2026
Office Agent: Unstructured Brief → Structured Plan with Approval Gates

WorkBuddy AI Platform

User
PM / Business Owner
Reviews & approves plans

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TOF Orchestrator
Main entry point | 8-step workflow
Manages sub-agents & approval gates

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Agent Architecture (6 Skills)

Planner
Brief → WBS
Effort & cost estimation
Risk identification

Validator
6 quality checks
Completeness, consistency
Feasibility, alignment

Executor
Executes approved actions
Writes to Excel tracker
Idempotency enforced

Quotation Gen
Plan → Quotation
Rate card calculation
Terbilang conversion

UI Designer
Plan → screen inventory
Every screen has derivedFrom
HTML → PNG mockups

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Deterministic Python Tools (8 modules, 157 self-tests)

gate.py
Approval state machine
27 self-tests
Prompt injection detection

validator.py
6-check plan validator
17 self-tests
PASS/WARN/FAIL verdict

quote_gen.py
Quotation generator
31 self-tests
Terbilang, payment schedule

ui_spec.py
Plan → screen inventory
33 self-tests
Every screen has derivedFrom

mockup_render.py
Spec → HTML → PNG
18 self-tests
5 layout archetypes

tracker_sync.py
plan.json → Excel
29 self-tests
Detects sheet-vs-plan drift

audit_logger.py
JSONL audit log
33 event types
Append-only

state_manager.py
JSON state management
Project CRUD
Approval lifecycle

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Hybrid Storage Architecture

Local JSON
Agent internal state
brief.json, plan.json
clarifications, approvals

Excel Tracker
Human-facing tracker
4 tabs, plan-verified
Dashboard + per project

Audit Log
JSONL daily files
33 event types
Append-only, traceable

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5 Accountability Rules (Hackathon Compliance)

Shows Sources
4 transparency tags:
BRIEF · CLARIFIED · ASSUMPTION · AI RECOMMENDATION

Within Permissions
Bounded authority
CAN analyze & propose
CANNOT execute w/o approval

Asks Before Acting
Approval state machine
DRAFT → PENDING → APPROVED
→ EXECUTED

Keeps Record
Audit trail: all events
Prompt injection detection
Completion honesty

Admits Limits
States what it cannot do
Flags 1 known test failure
No fabricated data

Approval State Machine (Reject → Remediate → Approve)

DRAFT → PENDING_APPROVAL → APPROVED → EXECUTED

REJECTED → DRAFT (revised) → PENDING (new) ← Remediation loop

FAILED (needs new approval to retry)